Mobile application Bluetooth frame documentation

February 18, 2021

1 Bluetooth data format

CANlink wireless series 3000 uses the RM Byte Command Protocol, which serves for bi-directional transparent CAN message transmission between the host and the CAN network. To transfer CAN message through Bluetooth it needs to be encoded into command frames with specific byte values.

Byte(s)	Value	Description
SOF (Start of Frame)	'C'	SOF marks the beginning of the command. 'C' (0x43) is the default setting. For some devices this value can be changed. This manual always shows the default setting.
Length	0x00-0xFF	The length byte includes the number of data bytes + the number of command bytes following.
Command	0x00-0xFF	Command byte, values 0x00 to 0xFE are direct commands, the value 0xFF means that the next byte contains a command extension
Data	xx	Data content
Checksum	xx	Contains the XOR checksum of the SOF, Length, Command and the Data-bytes.
EOF (End of Frame)	<CR>	EOF byte marks the end of the command. 0x0D (Carriage Return) is the default setting. For some devices this value can be changed. This manual always shows the default setting

"--" means that this value is not of interest
"xx" means that the field contains the configuration / reply value

For more info refer to RM Byte Command Protocol manual available in the projects Dropbox.

2 Sensor data info Bluetooth frames

2.1 Engine frame 1

From TTControl

- Accumulator voltage (ACC_VOL)
- Fuel Temperature (FT)
- Fuel Level (in litres) (FL)
- Engine Fuel Rate (EFR)
- Torque of engine (TRQ)
- Engine Rotations Per Minute (RPM)

Frame length = 8 bytes

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x64	RPM_1	RPM_0	TRQ	EFR	FL	FT	ACC_VOL_1	ACC_VOL_0

Decoding:

- $ACC_VOL = ((ACC_VOL_1 << 8)|(ACC_VOL_0))/1000$
- $EFR = EFR/10$

2.2 Engine frame 2

From TTControl

- ETH - engine total hours
- EFT - engine fuel total

Frame length = 8 bytes

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x65	ETH_3	ETH_2	ETH_1	ETH_0	EFT_3	EFT_2	EFT_1	EFT_0

Decoding:

- $ETH = ((ETH_1 << 8)|(ETH_0))/10$
- $EFT = ((EFT_1 << 8)|(EFT_0))/10$

2.3 Hydraulics data - nieaktualne, TODO(adam): uaktualnic

From TTControl

- Momentary predicted work time with remaining fuel (WT)
- Hydraulic oil temperature (OT - usint)
- Hydraulic oil pressure (PRS)

Frame length = 8 bytes

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x69	0xFF	WT	OT	PRS_1	PRS_0	RNG	0xFF	0xFF

Decoding:

- $OT = OT - 80$

TODO: add new hydraulics frames, figure out why in hal_mobile_app, ID of frame is equal to 0x6A

2.4 Carts angle and chassis angle

From TTControl

- CR_F - front cart rotation (CR=CR/10)
- CR_B - back cart rotation (CR=CR/10)
- X axis inclinometer value (along Agribot chasis), (X_INC=X_INC/10)
- Y axis inclinometer value (along Agribot chasis), (Y_INC=Y_INC/10)

Frame length = 8 bytes

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x6A	CRF_1	CRF_0	CRB_1	CRB_0	X_INC_1	X_INC_0	Y_INC_1	Y_INC_0

Decoding:

- All above values needs to be devided by 10 (same maner as in ETH in Frame 2).

3 State changing and command frames

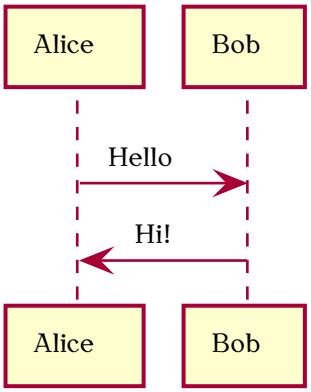
3.1 Heartbeat frame

From Mobile Application and TTControl. This frame needs to be send with proper value at least every 500 ms or robot will terminate the connection with mobile application. The initial Heartbeat value (HB_ID) of 0 is send by mobile application and then robot answers with the same value of HB_ID, which is then incremented by the application and send back. Robot once again check if the HB_ID is proper and send back the answer or breaks the connection if the value was bad. This process is repeated cyclically (at least every 200 ms), every 255 the HB_ID rounds back at 0.

Frame length = 1 byte

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x01	HB_ID	*	*	*	*	*	*	*

TODO: add sequence diagram



Example:

1. App send HB_ID = 0,
2. Robots checks if this is an initial value = 0 and sends it back,
3. App increment the value, and sends HB_ID = 1,
4. Robots checks if the difference between the received value the last one is equal to 1, its OK then sends it back,
5. ...
6. ...

3.2 Handshake frame

From Mobile Application. This frame initializes all the communication with the robot.

Frame length = 1 byte

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x0A	0x01	*	*	*	*	*	*	*

TODO: add sequence diagram of handshake procedure

TODO: add response frame 0x0b

TTControl send the exact same message as the confirmation of receiving this command.

3.3 Manual steering initialization

From Mobile Application. The data is only one dummy byte. This frame is used to initiate the manual steering state.

Frame length = 1 byte

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x14	0x14	*	*	*	*	*	*	*

TTControl send the exact same message as the confirmation of receiving this command.

3.4 Manual steering ready

From TTControl. The data is only one dummy byte. This frame is send to inform mobile application that the manual steering commands set is now active.

Frame length = 1 byte

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x15	0x15	*	*	*	*	*	*	*

TODO: add sequence diagram of manual steering initialization procedure

4 Error frames

4.1 Error frames

If the there is an error with robot, a frame describing the problem is send

- ERR – a 2-byte number describing error code

Frame length = 3 (?)

ID	B0	B1	B2	B3	B4	B5	B6	B7
0xC8	ERR_1	ERR_0	*	*	*	*	*	*

4.2 Engine error frames start

If there are errors in the engine, a frame announcing the problem is send

- ERR_TOT – a byte number describing total error count
- LAMP_STATE – the state of lamps which describe the severity of the problems

Frame length = 2 (?)

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x104	ERR_TOT	LAMP_STATE	*	*	*	*	*	*

4.3 Engine error frames

A frame containing one of the announced error frames. Byte order LSB first.

- ERR_NUM – an number number, value up to the announced in Engine error frames start
- SPN – the 3-byte error code described in Deutz documents
- FMI – the error specific code described in Deutz documents
- OCC – the error occurrence count
- ERR_STATE – if the error is active, contains 1, if passive 0

Frame length = 8

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x105	ERR_NUM	SPN_0	SPN_1	SPN_2	FMI	OCC	ERR_STATE	0xFF

5 Manual command frames

5.1 Override cart linear movement

Mobile application to TTControl.

- DIR - if 1, go forward, if 0, go backward
- CART_VEL - velocity of cart (in old agribot default value was 0xFF)
- FRONT_SEL - if 1, front cart selected. If 0, front cart deselected
- BACK_SEL - if 1, back cart selected. If 0, back cart deselected

Frame length = 4 (?)

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x32	DIR	CART_VEL	FRONT_SEL	FRONT_SEL	*	*	*	*

TODO: podobnie jak w Control tool position frame, wysylana jest ramka z ID: 0x32 i B0: 0x32. Dowiedziec sie dlaczego i czy jest to potrzebne

5.2 Override cart rotation

Mobile application to TTControl.

- CART_FRONT - if 1, front cart is selected and will rotate, if 0, front cart will be ignored
- DIR_FRONT - if 1, rotate front cart left, if 0, rotate front cart right
- CART_BACK - if 1, back cart is selected and will rotate, if 0, back cart will be ignored
- DIR_BACK - if 1, rotate back cart left, if 0, rotate back cart right

Frame length = 4 (?)

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x33	CART_FRONT	DIR_FRONT	CART_BACK	DIR_BACK	*	*	*	*

5.3 Joystick frame

From Mobile Application. This frame is used to move the robot around and needs to be send every 300 ms or the command timeout will occur and the controller will halt the outputs and the robot will stop.

- DIR_V - velocity direction (negative velocity means forward movement, positive - backward movement)
- DIR_R - radius direction (negative radius means right, positive - left)
- RAD - radius value
- VEL - velocity value

TODO: add radius (*10) and velocity values (*100) calculation

Frame length = 8

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x35	DIR_V	DIR_R	RAD_1	RAD_0	VEL_1	VEL_0	0xFF	0xFF

5.4 STOP frame

From Mobile Application. This frame is used to stop any ongoing track moving tasks like, aligning carts and switches. It is used to start and stop the engine too, by sending first this frame and then an Start Stop engine frame.

Frame length = 1

ID	B0	B1	B2	B3	B4	B5	B6	B7
0xFF	0xFF	*	*	*	*	*	*	*

TODO: add sequence diagram of stop engine procedure

5.5 Start/Stop engine frame

From Mobile Application. This frame is used to start the engine, by sending 2 of them in interval shorter than 200 ms and stop the engine by sending it for 2 seconds.

Frame length = 1

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x5A	0xFF	*	*	*	*	*	*	*

TODO: add sequence diagram of start engine procedure

5.6 Manual error frames

If the there is an error during manual commands, an error frame is sent

- ERR – a 2-byte number describing error code

Frame length = 2 (?)

ID	B0	B1	B2	B3	B4	B5	B6	B7
0xC9	ERR_1	ERR_0	*	*	*	*	*	*

6 Tools control

TODO: remove unnecessary config frames in hal_mobile_app

6.1 Control tool override - controlling the valves

From Mobile Application to TTControl (also from Advantech to TTControl). Every switch control a hydraulic valve connected to a specific part of the robot tools.

6.1.1 Frame

- SW_ID - switch ID number
- SW_ON - the switch should be ON (1) or OFF (0)
- SW_LV - velocity value a percent integer number from 0 to 100, numbers larger than 100 are treated as mistakes and the switch is instantly closed

Frame length = 3

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x3C	SW_ID	SW_ON	SW_LV	*	*	*	*	*

6.1.2 SW_ID

- 0x00 - Config to all switches
- 0x01 - FRONT_PTO
- 0x02 - FRONT_RED
- 0x03 - FRONT_GREEN
- 0x04 - FRONT_BLUE
- 0x05 - FRONT_TPL_ANGLE
- 0x06 - FRONT_TPL_HEIGHT
- 0x07 - BACK_PTO
- 0x08 - BACK_RED
- 0x09 - BACK_GREEN
- 0x0A - BACK_BLUE
- 0x0B - BACK_TPL_ANGLE
- 0x0C - BACK_TPL_HEIGHT

In return TTControl sends back confirmation frame:

Frame length = 1

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x3C	0x3C	*	*	*	*	*	*	*

6.2 Control tool position - manual mode steering

From Mobile Application to Advantech. In this case, the destination configuration of tool is sent via the CAN frame. The units inside the frame differs depending on SW_ID. Details are listed in below subsections.

6.2.1 Frame - set TPL position - request

- CONF_ID - config ID number (IDs are identical to those listed in section [6.1.2])
- CONF_VAL_Bx - values specific to ID

Frame length = 3

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x13D	CONF_ID	CONF_VAL_B1	CONF_VAL_B0	*	*	*	*	*

6.2.2 Frame - set TPL position response

From Advantech to Mobile app.

- CONF_ID - config ID number (IDs are identical to those listed in section [6.1.2])
- RET - 0xFF - failed to set TPL config (for instance values are out of range); 0x00 - successfully set TPL config

Frame length = 2

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x13E	CONF_ID	RET	*	*	*	*	*	*

6.2.3 CONF_VAL_Bx units and frame decoding

- FRONT_PTO\BACK_PTO - rotations per minute. Range from 0 RPM to 1000 RPM. The value in the frame is delivered on two bytes. The value can be calculated using following formula:
  $VAL = (CONF_VAL_B1 << 8)|(CONF_VAL_B0)$
- FRONT_GREEN\BACK_GREEN - valve flow control. Range from 0 to 100%.
  $VAL = (CONF_VAL_B0)$
- FRONT_YELLOW\BACK_YELLOW - valve flow control. Range from 0 to 100%.
  $VAL = (CONF_VAL_B0)$
- FRONT_BLUE\BACK_BLUE - valve flow from 0 to 100.
  $VAL = (CONF_VAL_B0)$
- FRONT_TPL_ANGLE\BACK_TPL_ANGLE - radians (from -0.26rad to 0.52rad). The negative number can be achieved by setting the highest bit to 1 in CONF_VAL_B1. The decoding can be performed by extracting sign bit, then setting the sign bit to 0. The resulding number should be devided by 1000.

  Example decoding: 0x8521. The 0x80 is indicates that the number is negative. We set the highest bit to 0. 0x0521 in hex is 1313. We divide the number by 1000 and multiply the number by -1. The result is -0.1313
- FRONT_TPL_HEIGHT\BACK_TPL_HEIGHT - ejection of prismatic joint (from 0.0m to 0.45m). The decoding process is analogous to to angle frame, but without the sign

6.3 TPL position feedback

Frame send periodically from Advantech to Mobile app. Feedback from sensors mounted on robot.

- CONF_ID - config ID number (IDs are identical to those listed in section [6.1.2])
- CONF_VAL_Bx - values specific to ID

Frame length = 3

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x13F	CONF_ID	CONF_VAL_B1	CONF_VAL_B0	*	*	*	*	*

7 RFID frames

7.1 Trigger scan

From mobile application to Advantech. The frame triggers scan from selected antenna.

- COM – a code number describing the meaning of the frame, in this case 1 (scan request)
- ANT_SEL – 0x01 - front antenna selected, 0x02 - back antenna selected, 0x03 - both antennas selected

Frame length = 2

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x1F4	COM	ANT_SEL	*	*	*	*	*	*

7.2 Detected tags - response

From Advantech to mobile application. In reply to trigger scan frame [7.1], the tags ID are sent. The EPC can be long up to 496 bits. In following example, the EPC tag is 9 bytes long.

- COM – a code number describing the meaning of the frame, in this case 2 (scan response start)
- TAG_ANT – 0x01 - tag detected by front antenna, 0x02 - tag detected by back antenna
- TAG_LEN – numbers of bytes
- TAG_ID – a 2-byte number representing the RFID tag ID

Frame length = 3

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x1F4	COM	TAG_ANT	TAG_LEN	*	*	*	*	*

Frame length = 8

- COM – a code number describing the meaning of the frame, in this case 3 (scan response tag data pt 1)

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x1F4	COM	TAG_ID_B0	TAG_ID_B1	TAG_ID_B2	TAG_ID_B3	TAG_ID_B4	TAG_ID_B5	TAG_ID_B6

Frame length = 8

If the ID is larger than capacity of frame, COM will increment.

- COM – a code number describing the meaning of the frame, in this case 4 (scan response tag data pt 2)

ID	B0	B1	B2	B3	B4	B5	B6	B7
0x1F4	COM	TAG_ID_B8	TAG_ID_B9	0x00	0x00	0x00	0x00	0x00